Chip in the Fast Lane

Nowadays, embedded systems are widely used as car computers. Embedded systems usually consist of a microprocessor along with ROM (Read Only Memory) as a single unit. The instructions stored in ROM are used to perform the specific operations of an embedded system. For example, a computer designed for engine control will be specifically designed with instructions in its memory to that end.

Embedded systems are preferred over all the other options because they can deal with large amounts of data from several sensors simultaneously. And they respond very quickly to data inputs. Consider what this means. The risk of brakes failing to work is considerably reduced if you use embedded systems. Besides, they are compact and cheap and can withstand the extreme conditions in a car. Another major advantage of using embedded systems is that they can be networked easily—all the computers controlling various processes can be connected using internal wiring to act as a single system.

Moving on to use of networked systems, there are individual computer modules for various functions such as engine control, safety, climate control, security, entertainment and so on. All these modules when networked together let you have single point control over all of them and their co-ordinated operation helps optimise the car's operation to higher levels of perfection.

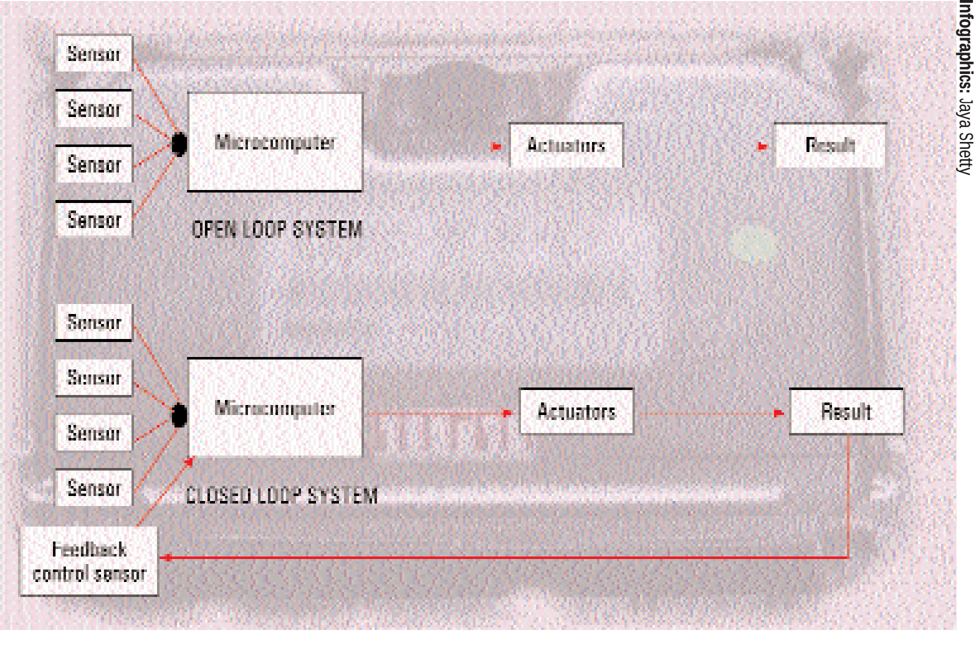
A Mind of its own

A single computer unit consists of several sensors connected to it. These sensors measure various parameters and send the values to the computer. The computer makes decisions and necessary action is taken using devices called actuators. Actuators convert electrical signals from the computer to mechanical actions. For example, if the window has to be opened, then the actuator does the mechanical actions for that. When you press the window button a signal is sent to the computer controlling it. The latter generates a signal, which is usually amplified and passed on to the motor, which converts the electrical signal into motion by opening the window. The motor acts as the actuator here.

There are two types of car computers—analog and digital. Nowadays, digital controllers are preferred because of their higher levels of accuracy. However, many signals in cars are in the analog form. This conversion from analog to digital form is done using devices called input conditioners. These input conditioners consist of analog to digital (A/D) converters and amplifiers to amplify the signal.

All the computers are interconnected using internal wiring.

There are two types of systems used in car computers—open loop and closed loop systems. Both of them are quite similar, the only difference being that in the closed loop system a feedback control sensor is used. In such an arrangement, the control unit gets to know whether the input it provided to the system has been applied correctly. If the output is not the desired one, then the controller can take corrective action. Closed loop systems are the most commonly used car computer configurations today as they have the advantage of monitoring the engine in a better fashion.



Infographics: Jaya Shetty

er of all used in a car as it has to handle a load of inputs from sensors situated in almost every section of the engine. When we say powerful computers, we are not talking about 1.4 GHz P4s or Athlons. Some of the most powerful computers used to control engines are most likely to be 32-bit processors running at approximately 40 MHz. However, unlike the conventional PCs we use, a processor used in these car computers do not have to deal with processing frills such as wallpapers or a dozen different applications running at the same time. All it has to do is carry out the instructions required to control a particular applications. For this, the speeds at which these ECUs operate are more than enough.

How exactly does an ECU work? In an engine (when you consider the mechanical aspect of it), there are sensors for sensing parameters such as throttle position, spark timing, transmission shift mode, vehicle speed, load, and so on. Other sensors commonly used in engines include sensors for oil pressure and temperature, and those for fuel and coolant temperatures. All these sensors are connected to the main ECU, which collects the data and takes decisions regarding their optimum settings. Most cars that use the fuel